Milestone #3
CompSci 325 - Fall 2025

Ctrl Park 🚗

Team Members

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Evaluation and Analysis

To complete an analysis of our UMass parking app prototype we went through a process of usability evaluations with unfamiliar users and cognitive walkthroughs. Across both methods we came up with a number of both quantitative and qualitative characteristics to understand possible issues regarding usability and assess how users interact with our design.

Evaluation Methods

Method 1: Usability Evaluation

To begin we conducted a usability study with 4 participants who were not a part of our group. We each engaged in one of these sessions, but used the same rubric across all instances and did not modify our prototype between to maintain consistency as much as we could. In order to make the most out of our testing, we ensured that participants were individuals who commuted or had a car on campus so they had real opinions on how effective the app was in comparison to what they had previously been exposed to. Each participant completed three tasks:

1. Reserve a parking space near their classroom
2. Modify the reservation
3. Navidate to the assigned lot and check in

During these sessions, we had participants verbalize their thoughts and actions so that we could fully understand what was going through their mind. As they completed the tasks we used a custom evaluation rubric to record several characteristics. These measures include:

- Time on Task
- App Navigation Method
- Misclicks, Confusion, Hesitation
- Accuracy of Task Completion
- Buttons Used

- Ease of Completion (scale of 1-5)
- Confidence in Completion (scale of 1-5)
- Behavior
- Narrative Comments

By evaluating these areas we were able to understand qualitative and quantitative insights on new users approaching our app. The full evaluation rubric is attached as Appendix A.

Method 2: Cognitive Walkthrough

After testing our app with unfamiliar users, we conducted a detailed cognitive walkthrough for the same three tasks. The process was modeled off of a walkthrough done by Chris Kimmer on YouTube, while simultaneously keeping in mind Interface Leanability as described by the Nelson Norman Group. For each step across the three tasks, we asked four standard cognitive walkthrough questions:

1. Will users try to achieve the right result?
2. Will users notice that the correct action is available?
3. Will users associate the correct action with the result they're trying to achieve?
4. After the action is performed, will users see that progress is made toward the goal?

By doing our own detailed evaluation, we were able to critique the interface of our app on an individual screen by screen basis and understand the flaws in its layout. Specifically, we could uncover unclear labeling and possible confusion between expectations and results within our prototype. The entirety of our cognitive walkthrough is attached as Appendix B.

Analysis

Once our prototype was tested using both methods, we mixed quantitative and qualitative analysis to come up with a holistic evaluation of our user interface usability given the three assigned tasks. With a small sample size we aimed to observe patterns in the usability sessions and cognitive walkthroughs rather than discover how statistically significant differences were. On the qualitative side, we completed a thematic analysis across common words, phrases, and issues experienced to summarize the strengths and weaknesses we discovered.

Quantitative Analysis

Out of the usability evaluations we were able to gather quantitative data on each of the three tasks participants were asked to complete. These statistics are as follows:

1. Task 1: Make a Reservation
    a. Minimum Completion Time: 51 seconds
    b. Maximum Completion Time: 4 minutes 30 seconds
    c. Average Completion Time: 2 minutes 10 seconds
    d. Average Ease of Completion: 3.1 / 5
    e. Average Confidence in Task: 4.5 / 5

2. Task 2: Modify Reservation (+30 minutes)
    a. Minimum Completion Time: 1 minute 3 seconds
    b. Maximum Completion Time: 4 minutes 30 seconds
    c. Average Completion Time: 2 minutes 15 seconds
    d. Average Ease of Completion: 2.9 / 5
    e. Average Confidence in Task: 3 / 5
3. Task 3: Navigate to Parking Lot and Check-In
    a. Minimum Completion Time: 27 seconds
    b. Maximum Completion Time: 6 minutes
    c. Average Completion Time: 2 minutes 45 seconds
    d. Average Ease of Completion: 3.5 / 5
    e. Average Confidence in Task: 2.8 / 5

Qualitative Analysis

To analyze our evaluation for the qualitative measures we took comments and behaviors and came up with seven major themes of areas that struggle in our low-fidelity prototype. These themes and narrative examples of the issues are as follows:

1. Poor Icon Discoverability
    a. Feedback: "Filter icon is small and not obvious" and "List view button confusing"
    b. Cognitive walkthrough verified the filters possibly not being noticed and that users may not associate icons with functionality.
2. Confusing Time Selection Control
    a. Multiple users clicked" OK" without adjusting clock
    b. Users asked for text input and end-time selection rather than duration
3. Map to Reservation Disconnect
    a. Feedback: "Didn't know which lot I reserved", "Didn't realize check-in only lives in the reservations page" and "Map and reservation screens disconnect is confusing"
    b. Cognitive walkthrough also backs these ideas as some people might not realize the lots are tappable and need direct prompts for certain actions
4. Visibility and Feedback Issues
    a. Feedback: "Need confirmation when reserving from lot view", "Did not realize one more step was needed (check-in)", and "Success screen clarity varies"
    b. Aligns with the negative feedback in Question 4 of the Cognitive walkthrough
5. Too Many Screens
    a. Feedback: "A lot of screens", "Too many steps", and "Busy layout"
    b. For the reservation functionality, there were a lot of steps as shown in Cognitive walkthrough
6. Unclear Interactive Elements

   a. Some users were hesitant to click on different buttons and had delayed actions spending more time thinking rather than deciding on what to click on
   b. Confirmed by the lack of noticing on tapable buttons
7. Want for Simplified Task Flows
   a. Users suggested a unified reservation list with options appearing on the same screen
   b. This appeared in two separate evaluations and was also backed by the drawn out tasks in reservation cognitive walkthrough

Findings

In both the usability evaluations and cognitive walkthrough, we came up with consistent themes of issues. Users struggled the most with discoverability and clarity, they were unclear with the filter icon, reservation time controls, and the map and reservation flow confusion. Quantitatively, we saw a large variability of task times, and an average low ease of completion scores. We also learned that the task flow was far too drawn out and confusing from comments from our participants. Overall, both analyses have led us to improve simplicity of task flows and the clarity of the user interface.

High-Fidelity Prototype

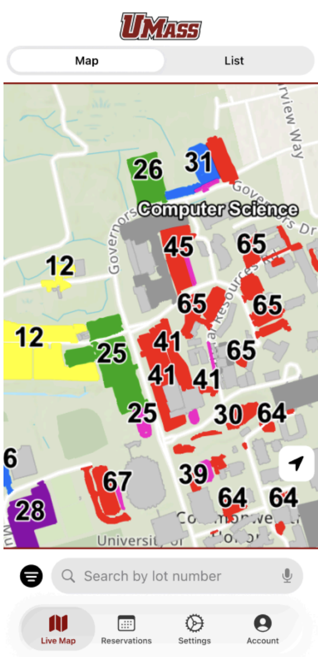
- What are the three primary features and tasks you implemented to address this problem?
  - How does your hi-fidelity solution differ from your proposal in milestone 1?
  - How does your hi-fidelity solution differ from your low-fidelity solution?
- How did the low fidelity prototype inform the design of the high-fidelity prototype?
- What parts were borrowed from the low-fi prototype? Did you learn something during the low-fi exploration in milestone 2?
- What challenges did you face while designing and developing your solution?
  - How did you address these challenges?

To create our high fidelity prototypes we used the analysis from both the cognitive walkthrough and the usability evaluation to identify issues in our original design. From there we were able to identify issues and correct them for the high fidelity prototype. This prototype we were able to click through and ensure that there was a solution to our identified problems. These problems were listed above in the qualitative analysis section. To correct these issues we…

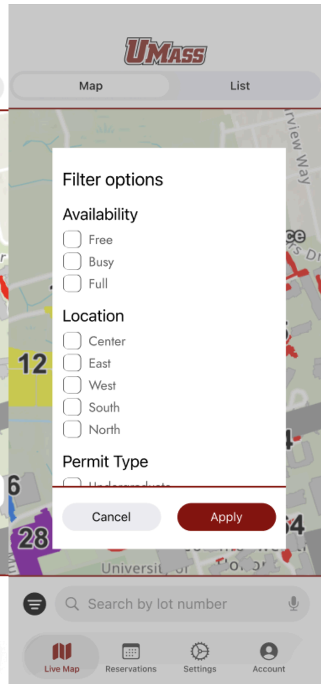
Live Map View

The live map view

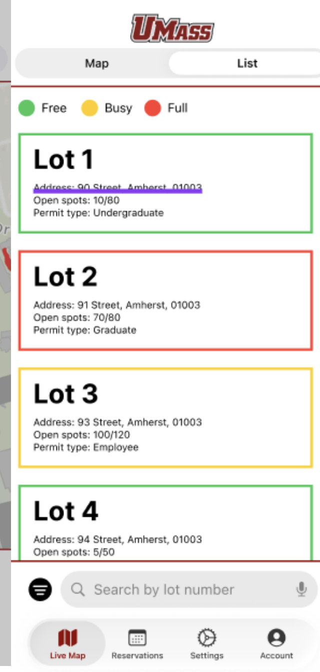
Map View



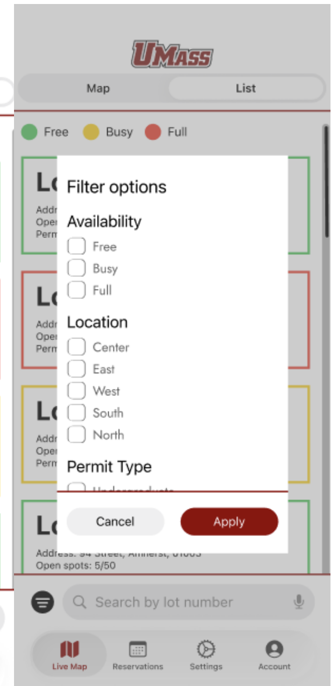
Map View - Filter



List View



List View - Filter

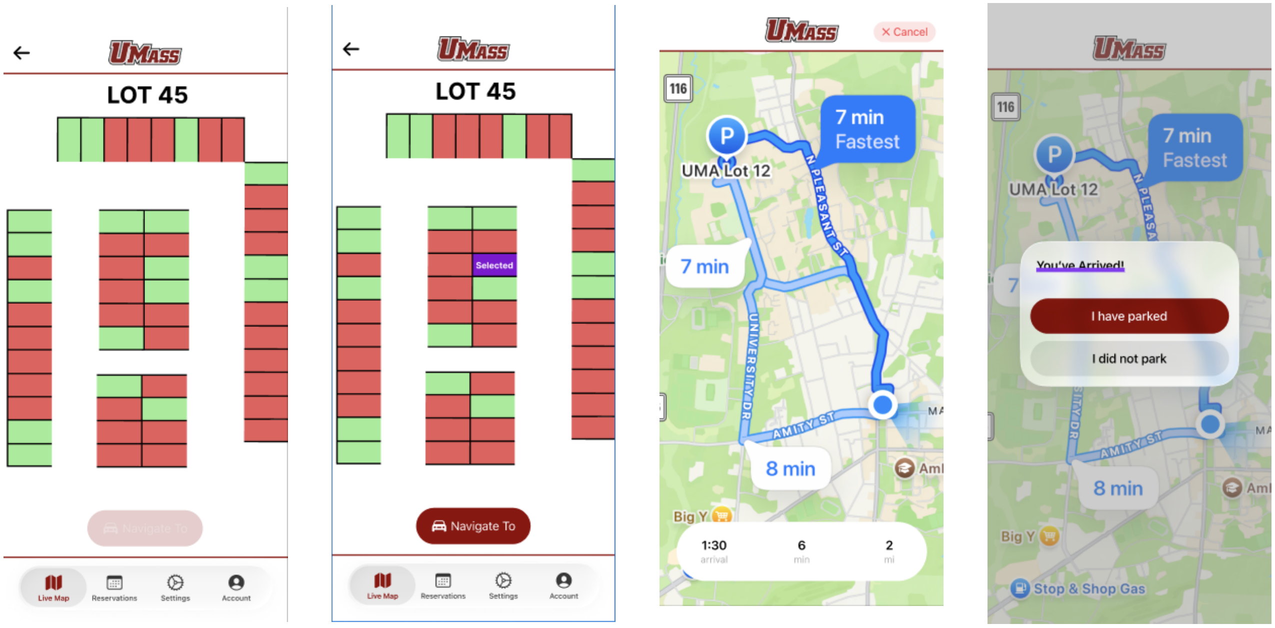


Lot View - Unselected

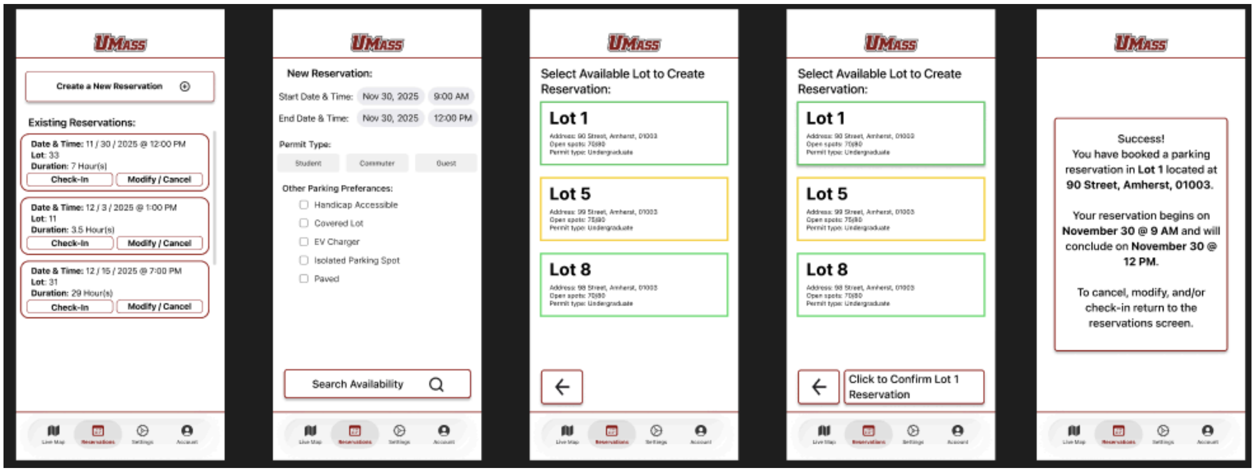
Lot View - Selected

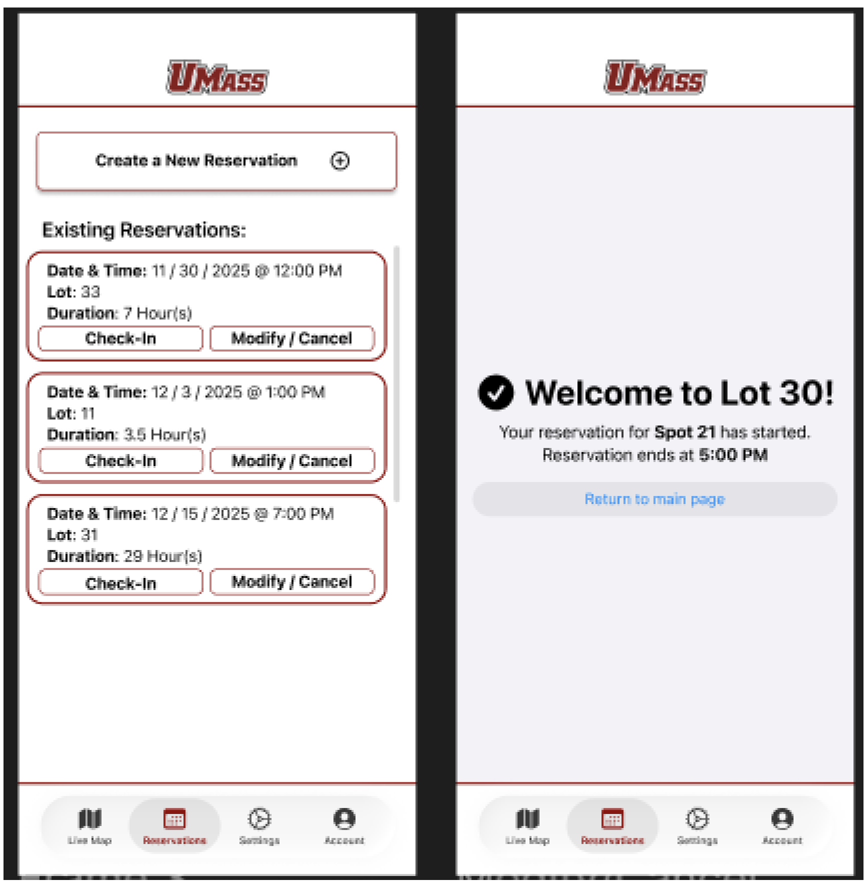
Navigate View

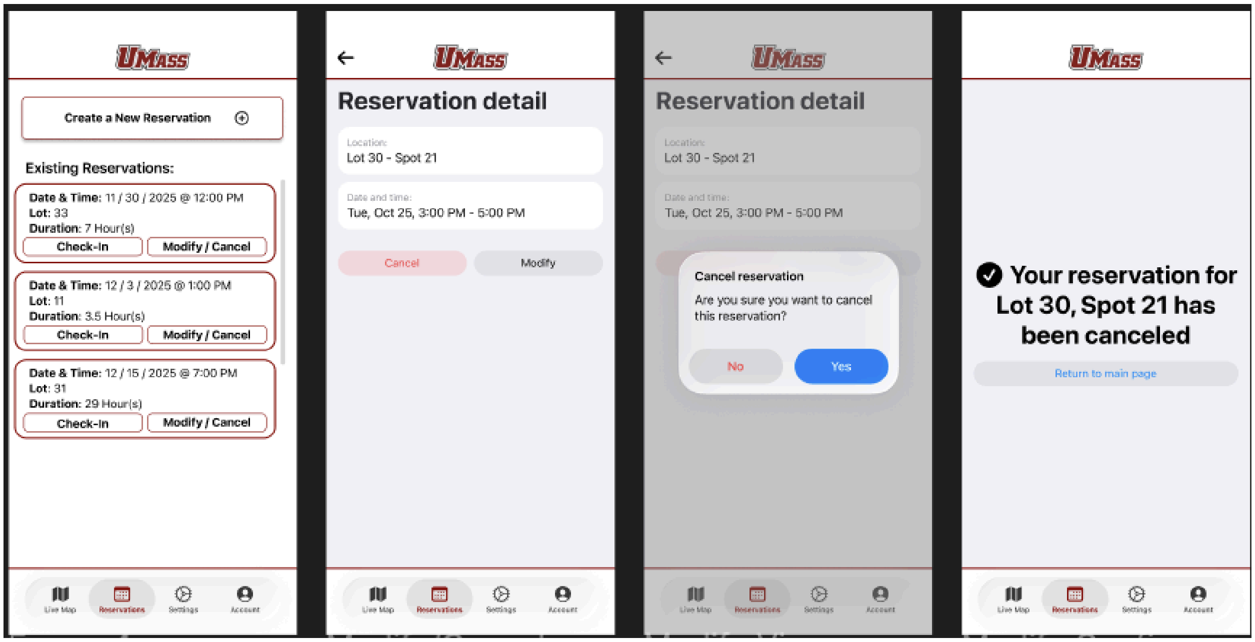
Navigate View Arrived

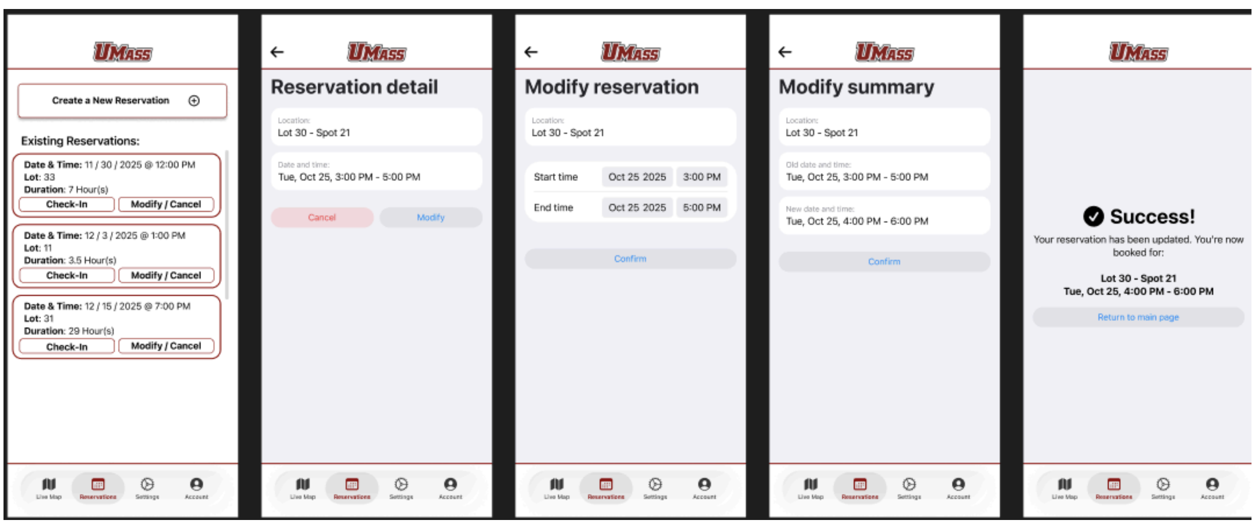

LOT 45
Navigate To
Live Map
Reservations
Settings
Account
LOT 45
Selected
Navigate To
Live Map
Reservations
Settings
Account
Cancel
116
7 min
Fastest
UMA Lot 12
N PLEASANT ST
7 min
UNIVERSITY DR
AMITY ST
8 min
Big Y
1:30
arrival
6
min
2
mi
116
7 min
Fastest
UMA Lot 12
You've Arrived!
I have parked
I did not park
8 min
Big Y
Stop & Shop Gas

Reservations Menu

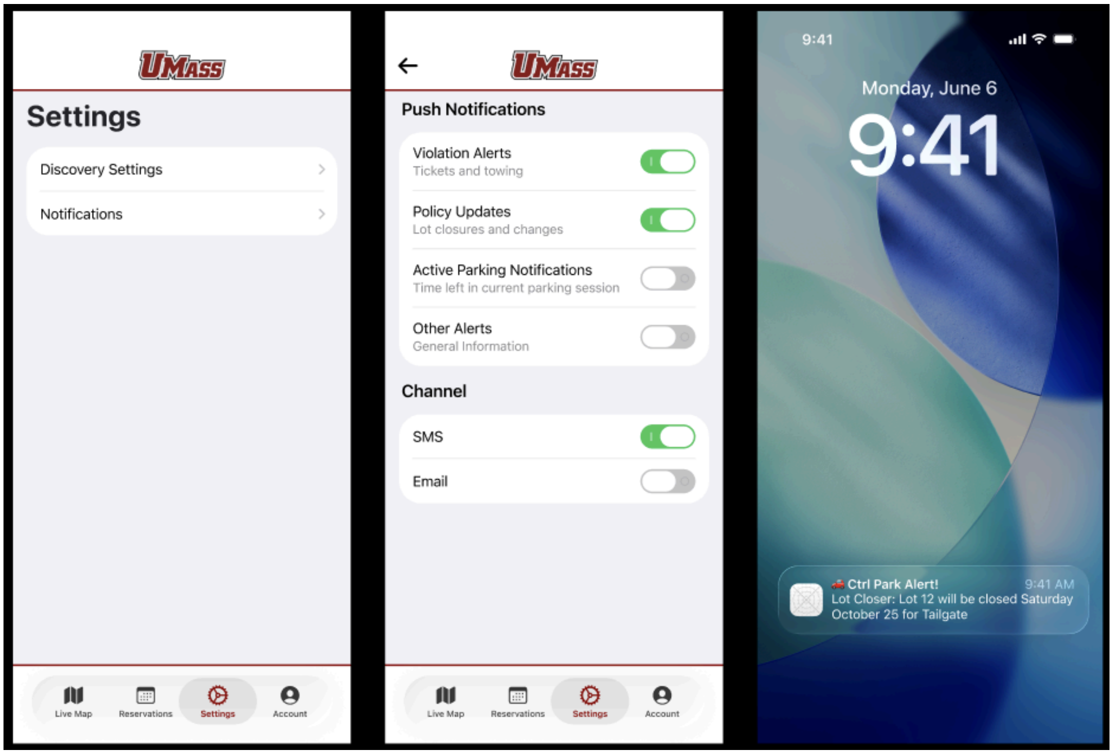


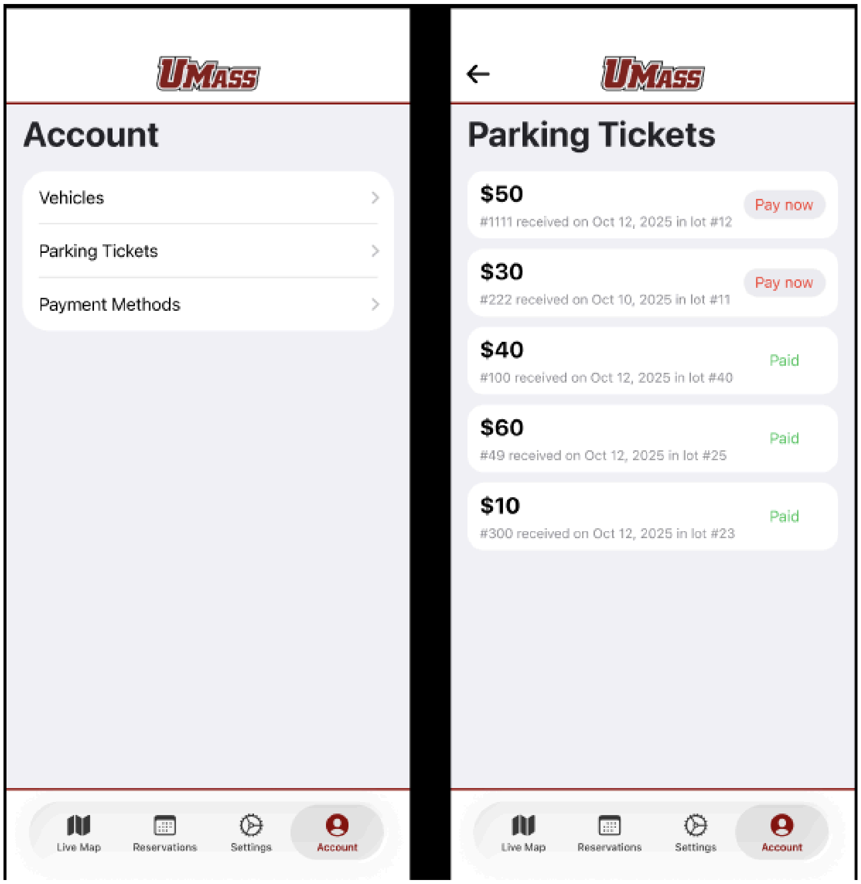


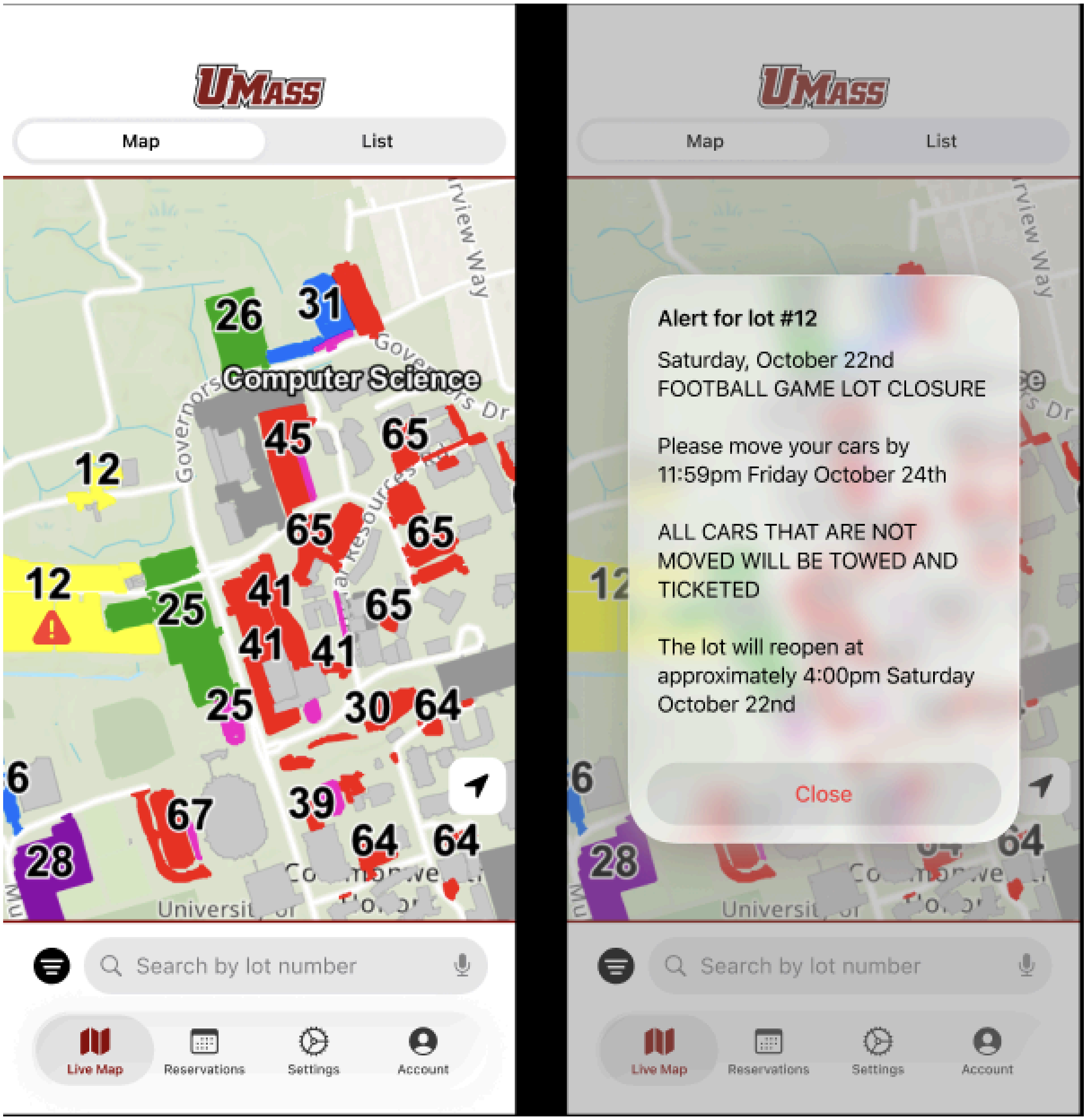

UMass
Create a New Reservation
Existing Reservations:
Date & Time: 11 / 30 / 2025 @ 12:00 PM
Lot: 33
Duration: 7 Hour(s)
Check-In
Modify / Cancel
Date & Time: 12 / 3 / 2025 @ 1:00 PM
Lot: 11
Duration: 3.5 Hour(s)
Date & Time: 12 / 15 / 2025 @ 7:00 PM
Lot: 31
Duration: 29 Hour(s)
Reservation detail
Location:
Lot 30 - Spot 21
Date and time:
Tue, Oct 25, 3:00 PM - 5:00 PM
Cancel
Modify
Cancel reservation
Are you sure you want to cancel this reservation?
No
Yes
Your reservation for Lot 30, Spot 21 has been canceled
Return to main page
Live Map
Reservations
Settings
Account

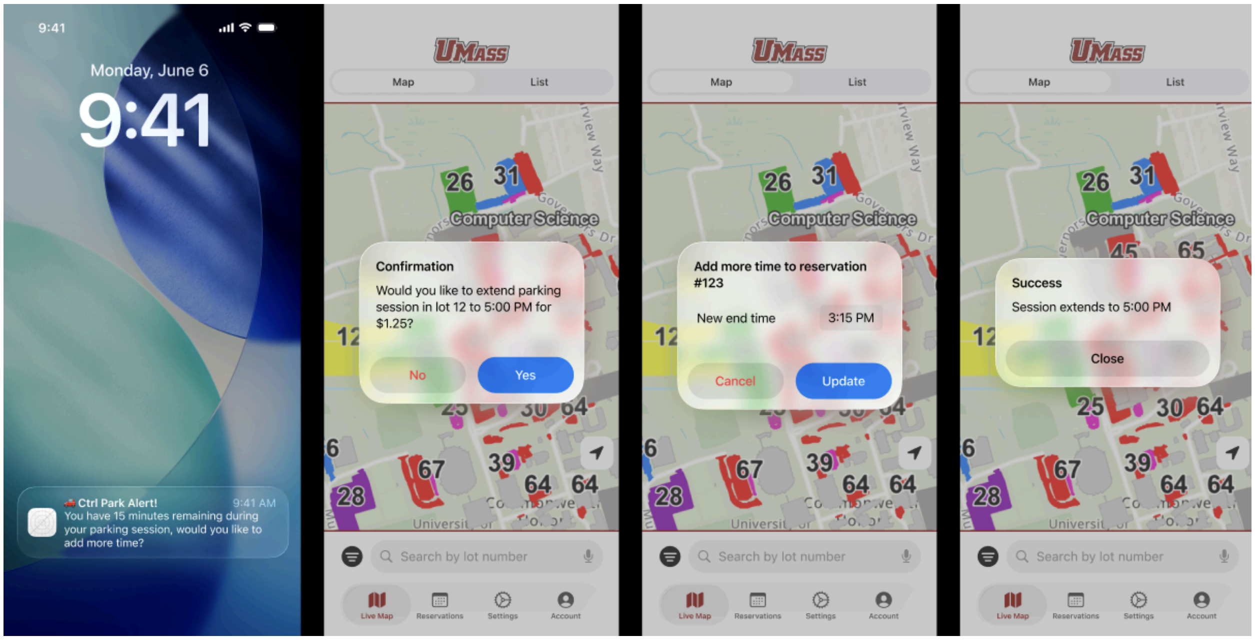

UMass
Create a New Reservation
Existing Reservations:
Reservation detail
Modify reservation
Start time
Oct 25 2025
3:00 PM
End time
5:00 PM
Confirm
Modify summary
Old date and time:
Tue, Oct 25, 3:00 PM - 5:00 PM
New date and time:
Tue, Oct 25, 4:00 PM - 6:00 PM
Success!
Your reservation has been updated. You're now booked for:
Lot 30 - Spot 21
Tue, Oct 25, 4:00 PM - 6:00 PM
Return to main page

Notifications and Violations




UMass
Account
Vehicles
Parking Tickets
Payment Methods
Live Map
Reservations
Settings
Account
UMass
Parking Tickets
$50
#1111 received on Oct 12, 2025 in lot #12
Pay now
$30
#222 received on Oct 10, 2025 in lot #11
Pay now
$40
#100 received on Oct 12, 2025 in lot #40
Paid
$60
#49 received on Oct 12, 2025 in lot #25
Paid
$10
#300 received on Oct 12, 2025 in lot #23
Paid
Live Map
Reservations
Settings
Account


UMass
Map
List
Computer Science
Governors Dr
Search by lot number
Live Map
Reservations
Settings
Account
Alert for lot #12
Saturday, October 22nd
FOOTBALL GAME LOT CLOSURE
Please move your cars by
11:59pm Friday October 24th
ALL CARS THAT ARE NOT
MOVED WILL BE TOWED AND
TICKETED
The lot will reopen at
approximately 4:00pm Saturday
October 22nd
Close



Participation

Erik: Created rubric for cognitive walkthrough. Completed a cognitive walkthrough with a UMass student outside the group. Created the map screen, list screen, and parking lot screen for the high fidelity prototype.

Kaylin: Completed a usability evaluation with a UMass student outside of the group. Completed a cognitive walkthrough for the “Reserve a Parking Spot” task. Completed a cognitive walkthrough for the “Turn on Notifications” task. Completed the analysis and write-up for both usability evaluation and cognitive walkthroughs. Created the High-fidelity prototype for “Creating a Reservation” via Figma.

Nam: Conducted two usability evaluations. Completed a cognitive walkthrough for “find parking spot and navigate to it” task. Created high-fidelity prototypes for map view, list view, navigation view, notification view, settings view, and account view.

Louise:

Appendix A: Usability Evaluation Rubric

The evaluation tasks depict an average user's natural daily usage of the app. Not based on what exactly the app can do, but what a user would expect and want to do with the app.

Context for user: You are a UMass student, and your class today is on the North side of campus at 11.30 AM, and ends at 12.45 PM.

Task 1: Reserve a parking space near your classroom for your class schedule.

Used map view, list view, or reservations menu?
Used filters?

Task 2: Your group partner texts you if you can stay an extra 30 minutes after class to work on your project. Modify your reservation accordingly

Task 3: It's time to head out, navigate to the parking spot. Once you arrive, check in.

****IT IS NECESSARY THAT THE PARTICIPANT THINKS ALOUD****

Task 1: Make A Reservation

Total Time Taken: ________

a) **Navigation Path Chosen (circle):** Map / List / Reservation Menu

b) Filter Usage (check all that apply)

1. Didn't use filters
2. Struggled to locate filter
3. Hesitated when selecting filters (took too long, backtracked, misclicked)
4. Incorrectly applied filters
5. Effectively used filters

Ease of Completion (1–5): ___

Notes/Comments:

c) Reservation Accuracy:

- ☐ Reserved for correct time (11:30–12:45)
- ☐ Reserved near correct location (North campus)
- ☐ Wrong location or time selected
- ☐ Unclear whether reservation succeeded

Ease of Completion (1–5): ___

Notes/Comments:

d) Feedback Recognition:

- ☐ Did not notice confirmation
- ☐ Unsure if reservation succeeded
- ☐ Recognized success screen immediately
- ☐ Commented positively on feedback clarity

e) Observed Behavior:

- ☐ Explored multiple views (Map/List)
- ☐ Stayed on one screen entire time
- ☐ Asked clarifying questions
- ☐ Commented on design/layout

Ease of Task (circle one): 1 2 3 4 5

Confidence in Task Completion (circle one): 1 2 3 4 5

Additional Feedback:

Task 2: Modify Reservation (+30 Minutes)

Total Time Taken: __________

a) **Navigation Effectiveness (check all that apply)**
   - ☐ Struggled to find Reservation Menu
   - ☐ Initially clicked Cancel by mistake
   - ☐ Opened wrong reservation first
   - ☐ Understood and navigated confidently

   Ease of Completion (1–5): ___

   Notes/Comments:

b) **Making Modifications (check all that apply)**
   - ☐ Incorrect changes
   - ☐ Unsure/hesitant when making changes
   - ☐ Correctly made changes
   - ☐ Confidently made changes

   Ease of Completion (1–5): ___

   Notes/Comments:

Ease of Task (circle one): 1 2 3 4 5

Confidence in Task Completion (circle one): 1 2 3 4 5

Additional Feedback:

Task 3: Navigate to Parking Lot, Check in once arrived

Total Time Taken: ________

a) **Navigation Effectiveness (check all that apply)**
   - ☐ Tried to use the search function on map screen
   - ☐ Clicked on the parking lot manually on the map screen
   - ☐ Went through multiple screens
   - ☐ Used the “Reservations” screen

   Ease of Completion (1–5): ___

   Notes/Comments:

b) **Start Navigation to parking lot (check all that apply)**
   - ☐ Struggled to start navigation
   - ☐ Couldn’t start navigation
   - ☐ Started navigation through the “Reservations” Screen
   - ☐ Started navigation through the “Map” screen

   Ease of Completion (1–5): ___

   Notes/Comments:

c) **Checking in once arrived (check all that apply)**
   - ☐ Forgot to check in
   - ☐ Struggled to spot the check in button
   - ☐ Knew immediately how to check in

   Ease of Completion (1–5): ___

   Notes/Comments:

Ease of Task (circle one): 1 2 3 4 5

Confidence in Task Completion (circle one): 1 2 3 4 5

Additional Feedback:

____

____

Appendix B: Cognitive Walkthrough

Context: You are a UMass student, and your class today is on the North side of campus at 11.30 AM, and ends at 12.45 PM.

Analysis Questions:

1. Will users try to achieve the right result?
2. Will users notice that the correct action is available?
3. Will users associate the correct action with the result they're trying to achieve?
4. After the action is performed, will users see that progress is made toward the goal?

Task: Find an available parking spot and navigate to it
Walkthrough:

1. The user clicks on the filter icon.
    - Question 1: **Yes.** A user who is unfamiliar with campus parking rules will likely want to narrow down the displayed lots to match their needs.
    - Question 2: **Yes.** The filter icon is present on the bottom navigation bar. However, its placement may cause some users to overlook its connection to the map, since bottom-tab actions are often perceived as global rather than map-specific. Relocating the icon onto the map interface (similar to the "center location" button) could improve visibility and association.
    - Question 3: **Yes.** The filter icon uses a conventional funnel symbol that is widely recognized by college-aged users as representing filtering tools. The main potential confusion is whether the filter applies to the map specifically or to the app overall, due to its current placement. Moving it into the map context could strengthen this association.
    - Question 4: **Yes.** A dedicated filter page appears at the center of the screen with a clear header indicating that filtering is being performed. To improve clarity, changing the header to "Filter Parking Lots" would make the purpose of the page even more explicit.
2. User selects "North" and "Undergraduate" filters
    - Question 1: **Yes.** Users intend to refine the list of parking lots by selecting relevant criteria such as location and permit type.
    - Question 2: **Yes.** The filter page clearly displays available categories (e.g., Location, Permit Types) accompanied by checkboxes.

    - Question 3: **Yes.** Checkboxes are a common UI control for selecting multiple options, and most users—especially college students—will understand their purpose.
    - Question 4: **Yes.** The checkboxes update visually with a checkmark. Clarity could be improved by displaying a summary count (e.g., “2 filters selected”).
3. User selects “Apply”
    - Question 1: **Yes.** After choosing criteria, applying the filters is a natural next step.
    - Question 2: **Yes.** The Apply button is visually distinct and consistently placed at the bottom of the filter page.
    - Question 3: **Yes.** The button is labeled “Apply,” which clearly communicates the intended function.
    - Question 4: **Yes.** The map updates to show only the matching parking lots, and the filter icon displays the number of active filters.
4. User clicks on a parking lot
    - Question 1: **Possibly.** Users may want to explore a specific lot, but they may not realize that lots on the map are interactive.
    - Question 2: **Yes.** Parking lots are bordered and color-coded, but this does not explicitly signal that they are interactive.
    - Question 3: **Possibly no.** Users may interpret the colored shapes as visual indicators, not interactive elements. Adding a legend explaining color meaning and a prompt such as “Tap a lot to view details” would improve clarity.
    - Question 4: **Yes.** Selecting a lot transitions to a view with its own map and labeled parking spots, along with a header naming the lot.
5. User selects desired parking space
    - Question 1: **Yes.** Users understand that they need to choose an available spot.
    - Question 2: **Yes.** Spots are bordered and colored, but numbering the spots could make them more identifiable.
    - Question 3: **No.** Users may not realize that individual spaces are interactive. An instruction such as “Select a parking space” would improve clarity.
    - Question 4: **Yes.** The selected spot is highlighted and marked as “Selected”, confirming the action.
6. User clicks “Navigate To”
    - Question 1: **Yes.** Once a spot is chosen, navigating to it is a logical next step.
    - Question 2: **Yes.** The button becomes enabled after a spot is selected.
    - Question 3: **Yes.** The button label “Navigate To” clearly communicates its purpose, though “Navigate to Spot” may be clearer.

- Question 4: **Yes.** The interface transitions to a navigation view showing directions to the selected spot.

7. User clicks “I Parked”
    - Question 1: **No.** Once parked, users may not expect to confirm the action in the app, since the necessity of tapping this button is unclear.
    - Question 2: **Yes.** The button is visible, but its importance may not be obvious. Enlarging or emphasizing it when users arrive at their spot could improve its visibility.
    - Question 3: **Yes.** The label “I Parked” is straightforward.
    - Question 4: **No.** There is currently no clear feedback indicating task completion. A confirmation page such as “You are parked at Lot X, Spot Y” would reinforce closure and reduce uncertainty.

Context: You are a UMass student, and your class today is on the North side of campus at 11.30 AM, and ends at 12.45 PM.

Analysis Questions:

5. Will users try to achieve the right result?
6. Will users notice that the correct action is available?
7. Will users associate the correct action with the result they’re trying to achieve?
8. After the action is performed, will users see that progress is made toward the goal?

Task: Reserve a parking spot
Walkthrough:

1. Click “reservations” option
    a. Question 1: **Yes**. button labeled “reservations” clearly indicated their goal.
    b. Question 2: Maybe. It is visible that the button is available, however the size of the button might lead people to look over it.
    c. Question 3: **Yes**. “reservations” would be a universal term to associate with reserving a spot
    d. Question 4: **Yes**. Users will be brought to a new screen with new options indicating progress.
2. Select “Create New Reservation”
    a. Question 1: **Yes**. Given that users want to book a new spot, they would naturally select “create new reservation”
    b. Question 2: **Yes**. The button is visible and separated from the other options with a clear label.
    c. Question 3: **Yes**. Selection describes the exact task they are hoping to accomplish.
    d. Question 4: **Yes**. After selection, the user is taken to a new screen, starting the reservation setup.
3. Choose Reservation day

    a. Question 1: **Yes**. Calendars align with the mental model of selecting a date.
    b. Question 2: **Yes**. Calendar is in the usual format and it is clear that it is a selectable interface.
    c. Question 3: **Maybe**. It is clear that tapping a date would select it, but they may not understand they need to move the clock dial in order to select a time.
    d. Question 4: Yes. Selecting options and clicking "OK" will move the user onto the next **screen**.
4. Choose duration, user type, other preferences
    a. Question 1: **Yes**. It is expected that a user would need to select their duration time for their reservation.
    b. Question 2: **Maybe**. While the duration field is visible, the preference buttons are small and the time formatting being different from the previous page may be confusing.
    c. Question 3: **No**. It is possible that a user may have totally overlooked the previous clock on another screen and be unsure if they are just reserving the spot for the duration at any point in the day.
    d. Question 4: **Yes**. Pressing "search availability" will bring the user to the next screen of lots and options.
5. View available lots
    a. Question 1: **Yes**. It is expected that the user would want to choose the lot they are reserving for.
    b. Question 2: **No**. Lots are simply colored blocks, it may not be clear they are physical buttons considering the small size of instruction text.
    c. Question 3: **Maybe**. Users familiar with interactive maps will find this intuitive, but others may not.
    d. Question 4: **Yes**. Upon click, the screen will switch to a detailed view, moving them along.
6. View and confirm lot details
    a. Question 1: **Yes**. Users will want to confirm their choice and understand all details of parking.
    b. Question 2: **Yes**. The "reserve" button is clear and visible.
    c. Question 3: **Yes**. It is also clear that the "reserve" button will have them confirming which lot they are reserving.
    d. Question 4: **Yes**. They will reach a new summary screen.
7. Review reservation summary
    a. Question 1: **Yes**. Once the lot is selected users will expect to finalize this decision.
    b. Question 2: **Yes**. The buttons on the bottom of the screen for "Confirm" and "Modify" clearly state the actions that they can take.
    c. Question 3: **Yes**. It is universal that "confirm" will complete the reservation.

d. Question 4: **Yes**. Users will be taken to the success screen. And understand that the reservation has been completed and actions are finished.

8. Success screen
    a. Question 1: **Yes**. There are no actions to be achieved but they will have finished the process.
    b. Question 2: **Yes**. Users will know the task is complete given the “Success!” text on the screen.
    c. Question 3: **Yes**. A success screen indicates completion.
    d. Question 4: **Yes**. The presentation of confirmation details indicates end of task flow.

Context: You are a UMass student and you use CtrlPark to do all of your on-campus parking, but need alerts sent to your phone.

Analysis Questions:

9. Will users try to achieve the right result?
10. Will users notice that the correct action is available?
11. Will users associate the correct action with the result they’re trying to achieve?
12. After the action is performed, will users see that progress is made toward the goal?

Task: Turn on app notifications to set up push notifications on all alerts.

1. Open Settings
    a. Question 1: **Yes.** We can assume that users will use the common settings gear to access notifications, although it may be located in a confusing spot for the app layout.
    b. Question 2: **Maybe**. The settings button is located next to Parking Availability and may seem like it has nothing to do with overall app settings.
    c. Question 3: **Yes**. Notifications are commonly associated with a setting.
    d. Question 4: **Yes.** They will be moved to a new menu that shows Notifications as one of the settings available through the app.
2. Select Notifications
    a. Question 1: **Yes**. It is clear that a user would click “Notifications” in order to get to actions regarding notifications.
    b. Question 2: **Yes**. The notifications button is clearly labeled and visible from the settings menu.
    c. Question 3: **Yes**. The term notifications match the mental model of where someone would go to find those options.
    d. Question 4: **Yes**. The user will jump to a new screen with alert categories separated by “Push Notifications” and “Channel”.
3. Turn on Notifications

a. Question 1: **Yes**. Users will recognize the toggles and alert types as ways to customize app notifications.
b. Question 2: **Yes**. The toggles align with the mental model of turning a switch on or off, similarly to notification status.
c. Question 3: **Maybe.** The toggles are clear, however the channel separation may not be as intuitive. They might not know to select both ways to get a notification.
d. Question 4: **Partially**. Users will see that the toggle will be set on, but there is no confirmation or test notification to signify that they are completely set up.